Geometric intuition in mathematics

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2025.05.18

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Overview

Differential geometry can be divided into classical differential geometry and modern differential geometry. The focus of differential geometry is curvature, which includes Gaussian curvature,sectional curvature, mean curvature and so on.

Algebraic topology primarily utilizes algebraic methods to describe topological invariants,such as the fundamental group and homology groups.

Knot theory belongs to low-dimensional topology,with a focus on finding invariants that are both computationally convenient and highly discriminative.These invariants can be numbers,polynomials,groups,etc.

Classical differential geometry

The Fundamental theorem of surface theory

The basic information of a surface embedded in three-dimensional Euclidean space is completely determined by the first fundamental form(E,F,G)and the second fundamental form(L,M,N).These forms are not entirely independent of each other and must satisfy a set of differential equations known as the Gauss-Codazzi equations.

The Gaussian theorema egregium

The Gaussian curvature at each point of a surface is completely determined by the three functions E,F,and G of the first fundamental form.

Hint: The metric on a surface also reflects the shape of the surface, and the higher-dimensional version of measuring shape with metrics is Riemannian geometry.

Classical differential geometry

Isoperimetric inequality

In the plane,for a simple regular closed curve with perimeter L,let S be the area enclosed by the curve.Then $S \leq \frac{L^2}{4\pi}$, where equality holds if and only if the curve is a circle.

Hint: We can associate this with knowledge related to the Riemann mapping theorem and fractal geometry.

Fenchel theorem

For a simple regular closed space curve C,its total curvature (the modulus of the derivative of the unit tangent vector) satisfies $\int_0^L k(s)ds \geq 2\pi, if and only if C$ is a simple regular convex closed plane curve.

Classical differential geometry

Fary-Milnor theorem

For a knotted simple regular closed curve,its total curvature satisfies the following inequality: $\int_0^L k(s)ds \geq 4\pi$.

Hint: The proof involves considering the method of contradiction and Crofton's formula on the sphere.The process is rather complicated and will not be elaborated here.

Liebmann theorem

In the 3-dimensional Euclidean space,a compact surface with constant curvature (Gaussian curvature , mean curvature) must be a sphere.

Hint: The precondition ensures that the Gaussian curvature is positive,from which we can deduce that it must be a sphere.

Algebraic topology

Fundamental group

The fundamental group is also known as the first homotopy group, and the most important concept within it is **homotopy**.To put it simply, homotopy is a continuous deformation between mappings, and the elements of the fundamental group are all the homotopy equivalence classes.

Basic facts

The fundamental group of a sphere is 0 , the fundamental group of a circle is $\mathbb{Z}$, and the fundamental group of a torus is $\mathbb{Z} \times \mathbb{Z}$, The fundamental group of the real projective space $\mathbb{RP}^n$ is $\mathbb{Z}/2\mathbb{Z}$ for $n \geq 2$, and for the complex projective space $\mathbb{CP}^n$, the fundamental group is trivial. The method for computing the fundamental group is usually the **Seifert–van Kampen theorem** (P8).

Algebraic topology

Homology group

Homology groups are algebraic invariants used in topology to describe the shape of a space.They capture information about the number of holes of different dimensions in a space. For example,the zeroth homology group counts **the number of connected components**, while higher homology groups describe **loops**(The abelianization of the fundamental group),**voids**, and **higher-dimensional holes** .

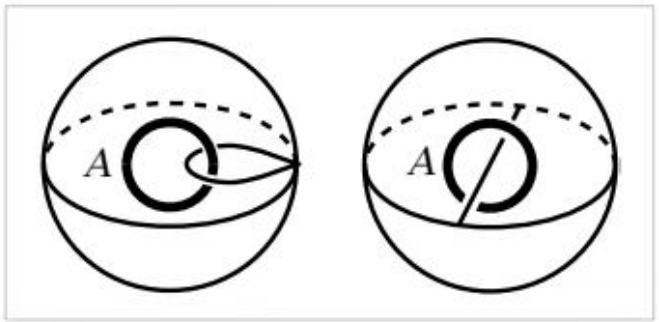
Hint: I will present the computation of the fundamental groups for some basic topological manifold, as well as the computation of the higher homotopy groups for the torus.

The geometric intuition of homotopy type

Let K is a subset of the 3-sphere $S^3 and be a smooth ly embedded circle$ (we call it a knot) in the 3-sphere S^3.
Compute all the homology groups of the complement $S^3 - K over \mathbb{Z}$.

Example 1.23: Linking of Circles. We can apply van Kampen's theorem to calculate the fundamental groups of three spaces discussed in the introduction to this chapter, the complements in $\mathbb{R}^3$ of a single circle, two unlinked circles, and two linked circles.

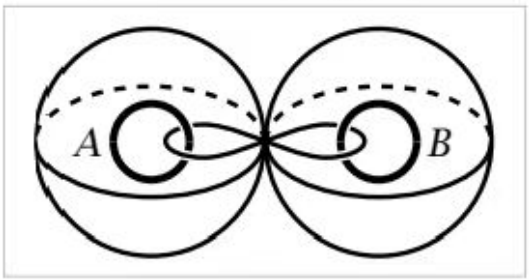
The complement $\mathbb{R}^3 - A$ of a single circle A deformation retracts onto a wedge sum $S^1 \vee S^2$ embedded in $\mathbb{R}^3 - A$ as shown in the first of the two figures at the right. It may be easier to see that $\mathbb{R}^3 - A$ deformation retracts onto the union of S^2 with a diameter, as in the second figure, where points outside S^2 deformation retract onto S^2, and points inside S^2 and not in A can be pushed away from A toward S^2 or the diameter. Having this deformation retraction in mind, one can then see how it must be modified if the two endpoints of the diameter are gradually moved toward each other along the equator until they coincide, forming the S^1 summand of $S^1 \vee S^2$. Another way of seeing the deformation

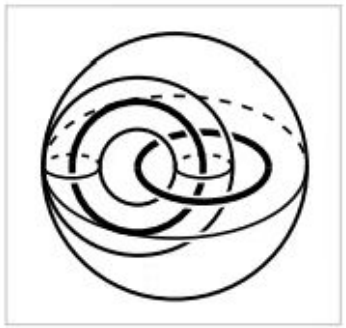


retraction of $\mathbb{R}^3 - A$ onto $S^1 \vee S^2$ is to note first that an open ε-neighborhood of $S^1 \vee S^2$ obviously deformation retracts onto $S^1 \vee S^2$ if ε is sufficiently small. Then observe that this neighborhood is homeomorphic to $\mathbb{R}^3 - A$ by a homeomorphism that is the identity on $S^1 \vee S^2$. In fact, the neighborhood can be gradually enlarged by homeomorphisms until it becomes all of $\mathbb{R}^3 - A$.

In any event, once we see that $\mathbb{R}^3 - A$ deformation retracts to $S^1 \vee S^2$, then we immediately obtain isomorphisms $\pi_1(\mathbb{R}^3 - A) \approx \pi_1(S^1 \vee S^2) \approx \mathbb{Z}$ since $\pi_1(S^2) = 0$.

In similar fashion, the complement $\mathbb{R}^3 - (A \cup B)$ of two unlinked circles A and B deformation retracts onto $S^1 \vee S^1 \vee S^2 \vee S^2$, as in the figure to the right. From this we get $\pi_1(\mathbb{R}^3 - (A \cup B)) \approx \mathbb{Z} * \mathbb{Z}$. On the other hand, if A and B are linked, then $\mathbb{R}^3 - (A \cup B)$ deformation retracts onto the wedge sum of S^2 and a torus $S^1 \times S^1$ separating A and B, as shown in the figure to the left, hence $\pi_1(\mathbb{R}^3 - (A \cup B)) \approx \pi_1(S^1 \times S^1) \approx \mathbb{Z} \times \mathbb{Z}$.





Homotopy groups of spheres

	π_1	π_2	π_3	π_4	π_5	π_6	π_7	π_8	π_9	π_{10}	π_{11}	π_{12}	π_{13}	π_{14}	π_{15}
S^0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^1	$\mathbb{Z}$	0	0	0	0	0	0	0	0	0	0	0	0	0	0
S^2	0	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12}\times\mathbb{Z}_2$	$\mathbb{Z}_{84}\times\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^3	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{12}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{12}\times\mathbb{Z}_2$	$\mathbb{Z}_{84}\times\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$
S^4	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}\times\mathbb{Z}_{12}$	$\mathbb{Z}_2^2$	$\mathbb{Z}_2^2$	$\mathbb{Z}_{24}\times\mathbb{Z}_3$	$\mathbb{Z}_{15}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{120}\times\mathbb{Z}_{12}\times\mathbb{Z}_2$	$\mathbb{Z}_{84}\times\mathbb{Z}_2^5$
S^5	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{30}$	$\mathbb{Z}_2$	$\mathbb{Z}_2^3$	$\mathbb{Z}_{72}\times\mathbb{Z}_2$
S^6	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_{60}$	$\mathbb{Z}_{24}\times\mathbb{Z}_2$	$\mathbb{Z}_2^3$
S^7	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}_{120}$	$\mathbb{Z}_2^3$
S^8	0	0	0	0	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_{24}$	0	0	$\mathbb{Z}_2$	$\mathbb{Z}\times\mathbb{Z}_{120}$

设 $\boldsymbol{H}$ 是四元数的集合, i, j, k 是它的生成元, 它们满足

$$i^2 = j^2 = k^2 = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j.$$

任一四元数可以表示为 $v_1 + v_2 i + v_3 j + v_4 k$, $v_i \in R$, 类似例 1 利用四元数的乘法, 可以构造 S^3 或者 S^{4k-1} 上三个处处线性无关的切向量场. 而利用 Cayley 数的乘法, 可以构造 S^7 上七个不同的切向量场. 一般奇数维的球面上非零切向量场的研究, 特别是它上面线性无关的非零切向量场的个数是一个有趣的问题. 利用 Clifford 代数及它们的不可约表示空间可以证明

定理 4.5.8 对于任一奇数 n, 记 $n+1 = 2^{4a+b}(2k+1)$, a, b, k 是非负整数, $0 \leqslant b \leqslant 3$, 那么在球面 S^n 上有

$$8a + 2^b - 1$$

个线性无关的非零切向量场.

如果 $n+1 = 2(2k+1)$, 球面 S^n 上只有 1 个非零切向量场. 下面是一些特例.

$n = 3$ 时, $4 = 2^2$, $a = 0, b = 2, k = 0$, 球面 S^3 上有 $2^b - 1 = 3$ 个线性无关的非零切向量场. 在 $n = 7$ 时, $8 = 2^3$, $a = 0, b = 3, k = 0$, 球面 S^7 上有 $2^b - 1 = 7$ 个线性无关的非零切向量场. 在 $n = 15$ 时, $16 = 2^4$, $a = 1, b = 0, k = 0$, 球面 S^{15} 上有 $8a + 2^b - 1 = 8$ 个线性无关的非零切向量场.

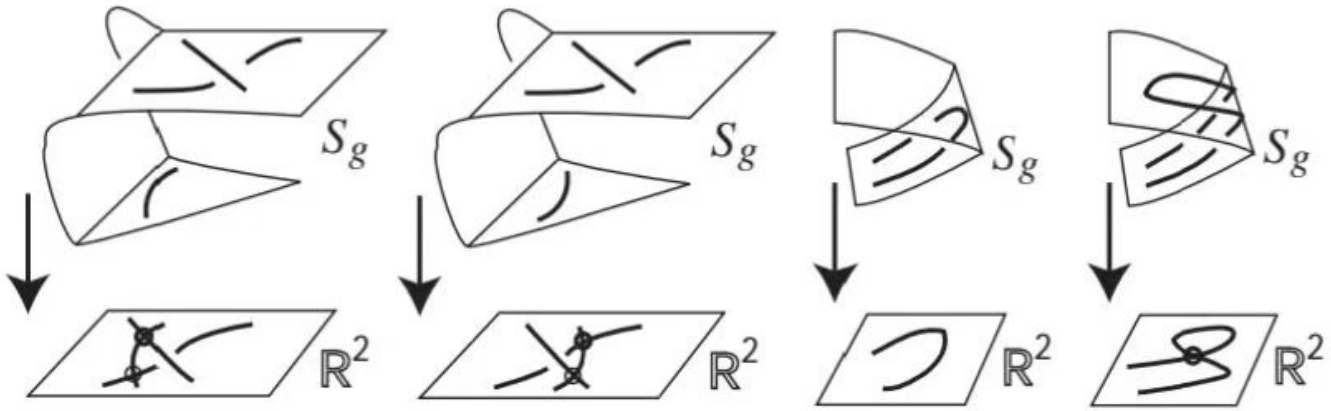
Adams 证明了 $8a + 2^b - 1$ 也是球面 S^n 上线性无关的非零切向量场的最大数, 在这一结论的证明中代数拓扑的知识起了关键作用.

Knot theory

Basic introduction

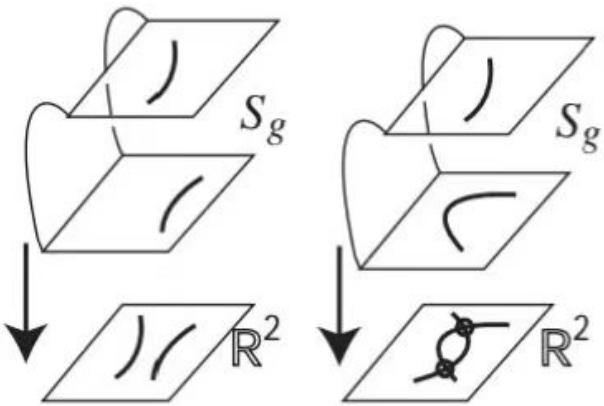
The core of knot theory is to find invariants that are both easy to compute and highly discriminative.Simply say, **a knot is a connected,closed,non-self-intersecting simple closed curve in three-dimensional space**.A circle placed in a plane is considered an"unknotted knot,"which we call the trivial knot.If one knot can be continuously deformed into another knot,we say that these two knots are isotopic.Knots that cannot be further continuously deformed are called prime knots.If there are more than one non-intersecting circles in a plane,we call it a trivial link.When studying knots,we usually use their projections on a plane,with the over-under relationships indicated by solid and dashed lines.

Reidemeister move

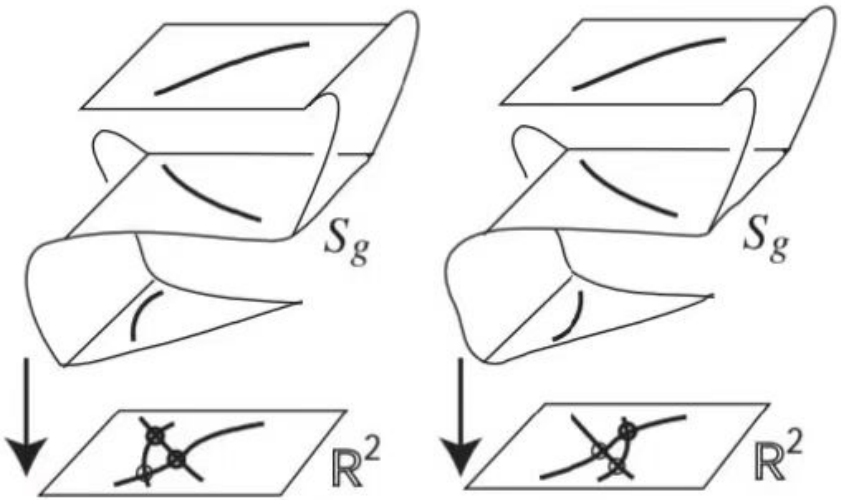


The semivirtual move

The first virtual Reidemeister move



The second virtual Reidemeister move



The third virtual Reidemeister move

The calculation of the Jones polynomial

如果有向投影图 L 与 L' 互相同痕，它们所对应的多项式相等，$V(L)=V(L')$.

(2)拆接关系式

$$t^{-1}\cdot V(\times)-t\cdot V(\times)=(t^{1/2}-t^{-1/2})\cdot V(\cup),$$

其中的 $\times$, $\times$, $\cup$ 代表三个几乎完全一样的有向投影图，只在某一个交叉点附近有画出的不同形状.

(3)标准值

平凡纽结 $\bigcirc$ 所对应的多项式是

$$V(\bigcirc)=1.$$

我们称 $V(L)$ 为 L 的琼斯多项式.

conclusion

Due to time constraints,I will stop the introduction to knot-related knowledge here.I warmly welcome everyone to come and discuss with me offline!

Thank you all very much for your attention!